Shoolini University

Final Project Report – Completion Stage

Submission Date: 25 November 2026

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1. Student Details

Name	Md Faiz Alam
Course	BCA (Bachelor of Computer Applications)
Semester	6th Semester
Admission No.	UGO24156735
Project Title	StockFlow – Inventory Management System for Small and Medium Businesses
GitHub	https://github.com/Md-Faiz-Alam/inventory-management-system
Live App	https://stockflow-inventory-vtqp.onrender.com

2. Project Title & Objective

Title: StockFlow – Inventory Management System for Small and Medium Businesses

Objective: StockFlow is a fully deployed, web-based inventory management system for small and medium businesses that still keep stock in registers or spreadsheets. It keeps an exact, real-time record of every product and every unit that comes in or goes out, with who made the change and why. On top of that record it manages suppliers, purchase orders, customers and sales orders, produces invoices and reports, and gives managers a live dashboard that shows what needs reordering – all accessible from any device through a web browser.

3. Final Features Implemented

Access and security

- Secure sign-in with four roles: Admin, Manager, Staff and read-only Viewer.
- One-click Explore as guest for visitors: every page can be viewed, nothing can be changed.
- User management: add users, change roles, deactivate accounts and set new passwords.
- Account lock for 15 minutes after five failed sign-ins (web page and API).

- Activity log of every create, edit, status change, stock operation, order step and sign-in, with old and new values and the IP address.
- HTTPS only, secure cookies, CSRF protection and browser security headers in production.

Inventory

- Product catalogue with unique SKUs, categories, prices and reorder levels; archive instead of delete.
- Receive and issue stock through one locked, transactional service; stock can never go negative and double submissions are ignored.
- Complete movement history; corrections by reversal or stocktake.
- Low-stock and out-of-stock indicators and a top-bar alert that refreshes every minute.

Purchasing and sales

- Suppliers linked to the products they provide.
- Purchase orders with partial deliveries, delivery document numbers, progress and cancellation.
- Customers and multi-line sales orders (Draft → Confirmed → Fulfilled); fulfilment issues all lines in one transaction or none.

Insights and documents

- Analytics dashboard: stock value, product and unit totals, low/out-of-stock counts, open purchase orders, monthly sales, plus charts of stock in vs. out (30 days), stock health, value by category and fastest-moving products.
- Stock and movement reports with filters, exported as CSV or PDF.
- PDF purchase orders and tax invoices / quotations.

Interface and integration

- Live search and filtering without page reloads (HTMX) on every list.
- Responsive design for desktop and mobile, with light and dark themes.
- Public landing page showing live figures from the database.
- REST API (products, suppliers, customers, stock movements, purchase and sales orders) with token sign-in and interactive Swagger documentation.

4. System Working Explanation

Architecture

StockFlow is a Django web application backed by PostgreSQL. Pages are rendered on the server and made interactive with HTMX and Chart.js; a REST API exposes the same data to other programs. Every stock change – from a page, a purchase order, a sales order or the API – goes through one service function that locks the product row, checks the rules and writes an unchangeable movement record, which keeps balances correct even when several users work at the same time.

Layer	Technology
Backend	Python 3.11, Django 5.2, Django REST Framework, drf-spectacular
Database	PostgreSQL 17 (Docker locally, Neon in production)
Frontend	HTML, CSS, JavaScript, HTMX, Chart.js
Documents	xhtml2pdf (PDF), CSV export
Testing and CI	Django test framework (217 tests), GitHub Actions
Hosting	Render (Gunicorn + WhiteNoise) and Neon PostgreSQL

Step-by-step working

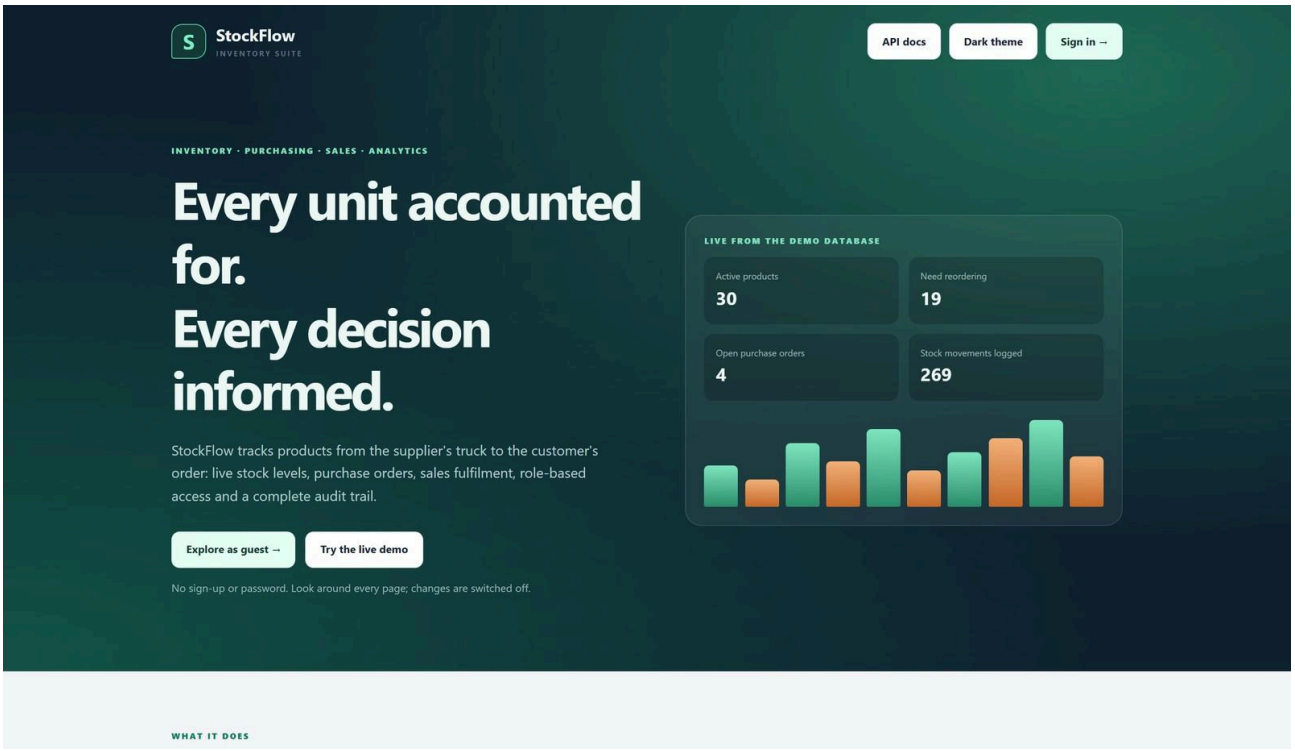
- Step 1 – Visit: the landing page shows what the system does and live totals. A visitor can click Explore as guest (read-only) or sign in.
- Step 2 – Sign in: the user's role decides which menu items and actions appear; every page also checks permissions on the server.
- Step 3 – Dashboard: key figures and four charts are loaded from the database; the alert bell shows how many products are at or below their reorder level.
- Step 4 – Catalogue: a Manager adds products and suppliers and links them; lists update instantly while typing in the search box.
- Step 5 – Purchase: from a low-stock product, the Manager raises a purchase order (supplier and cost are pre-filled) and can download it as a PDF.
- Step 6 – Receive: when goods arrive, Staff record the delivery against the order; stock rises and the order shows what is still outstanding.
- Step 7 – Sell: a sales order is created for a customer, confirmed by a Manager and fulfilled by Staff, which issues stock for every line; a tax invoice PDF is then available.

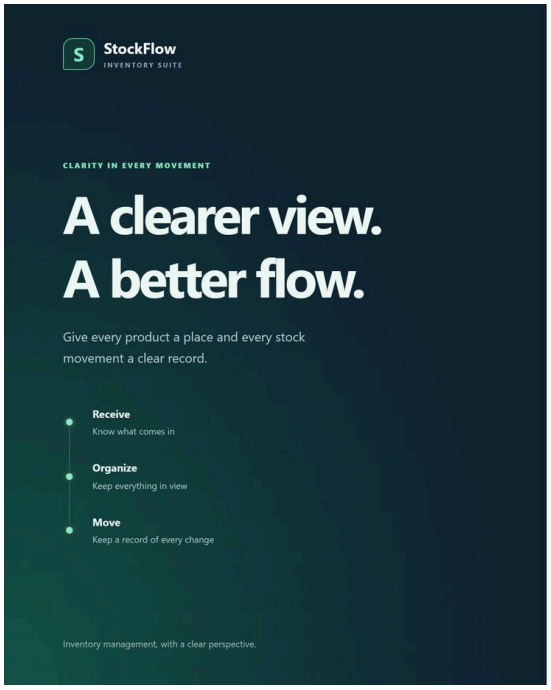
- Step 8 – Correct: mistakes are fixed by reversing a movement or by a stocktake, so history is never edited or lost.
- Step 9 – Review: reports (HTML, CSV, PDF), the movement history and the activity log show exactly what happened, when and by whom.

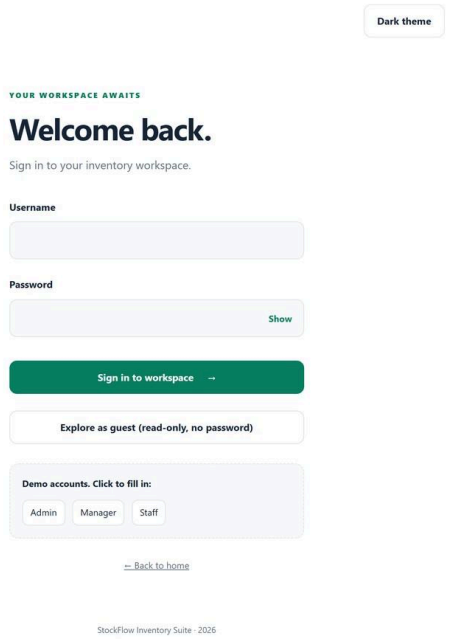
Testing and deployment

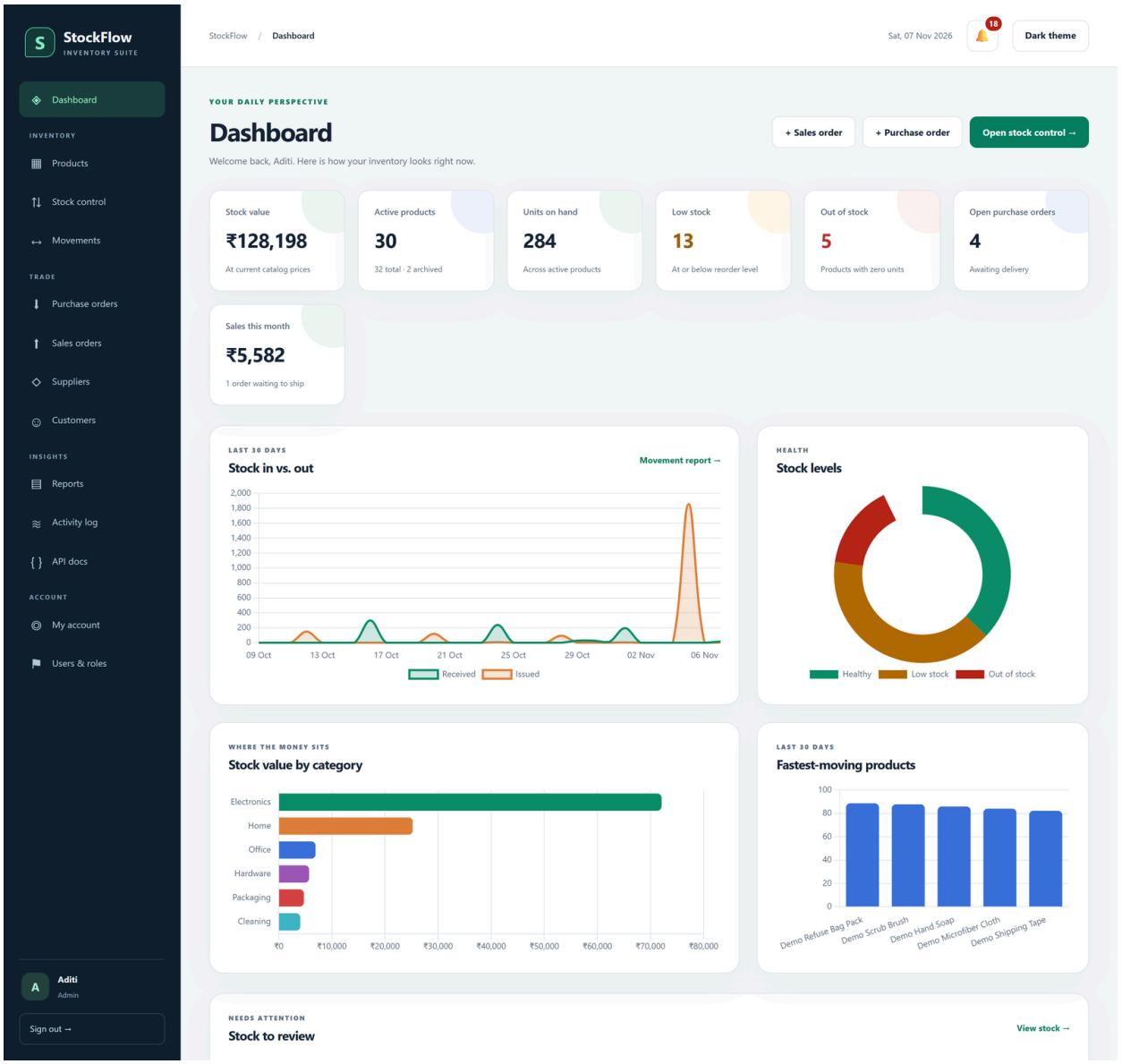
- 217 automated tests cover every module, including simultaneous stock operations, permissions for each role, the API, PDFs and the guest view.
- GitHub Actions runs on every push: tests on PostgreSQL 17, migration checks, demo data loading, production security checks and a server start-up test.
- A website checker script signs in like a visitor and opens every page, export, PDF and API endpoint. On the live site: 66 checks passed, 0 failed.
- Deployed on Render (free web service) with a Neon PostgreSQL database. Every push to GitHub redeploys automatically; HTTP is redirected to HTTPS.

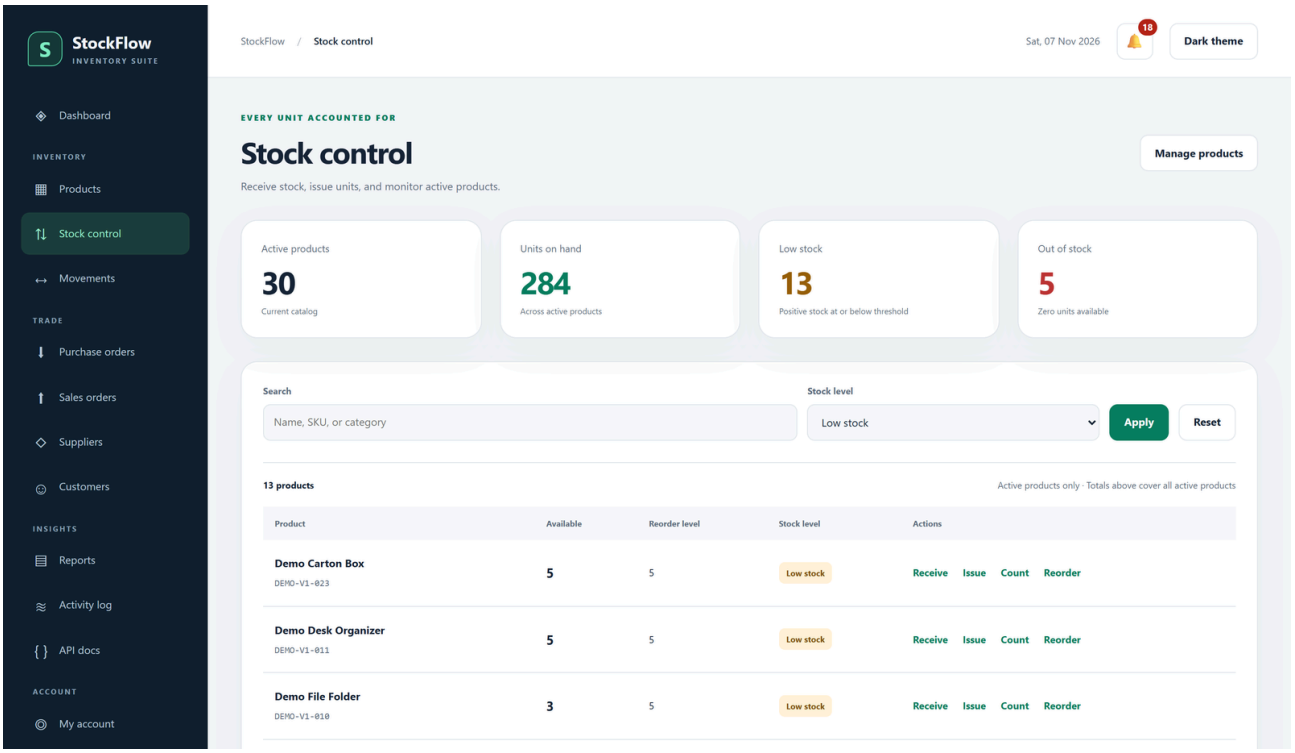
5. Screenshots / Output

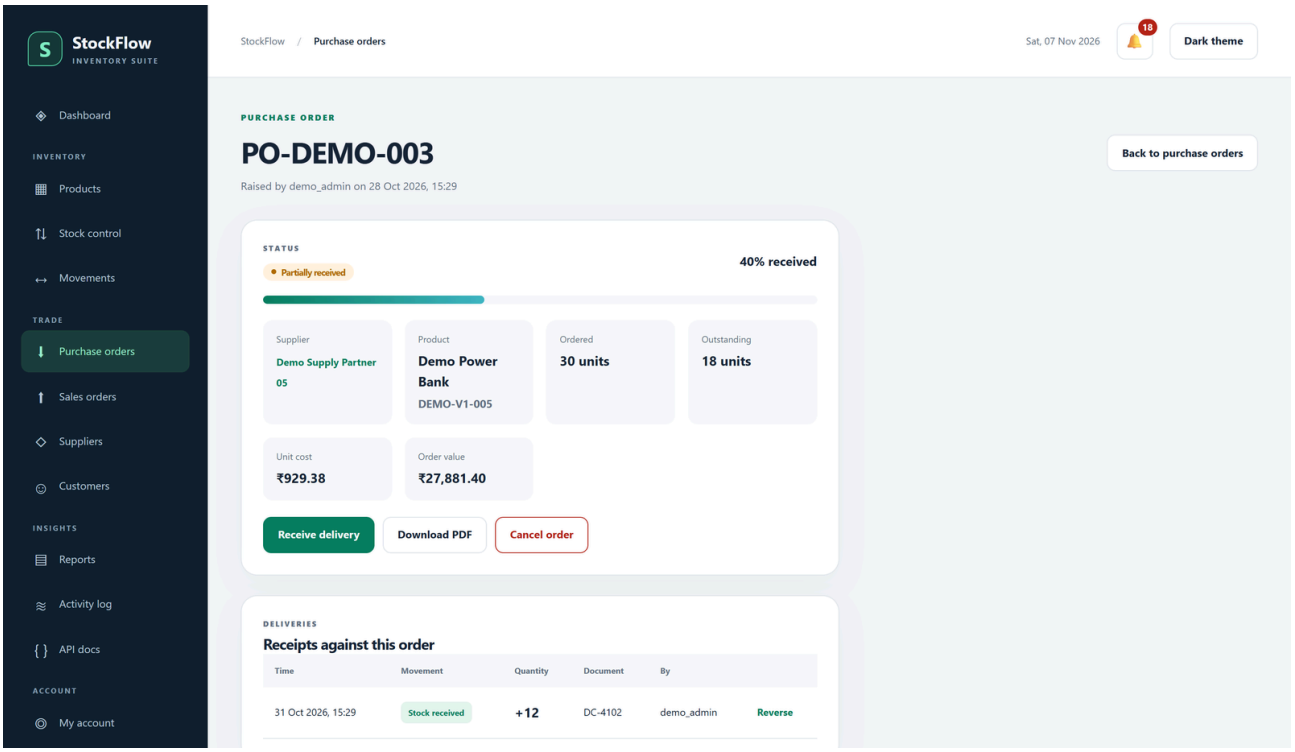



StockFlow
INVENTORY SUITE
CLARITY IN EVERY MOVEMENT
A clearer view.
A better flow.
Give every product a place and every stock movement a clear record.
Receive
Know what comes in
Organize
Keep everything in view
Move
Keep a record of every change
Inventory management, with a clear perspective.


Dark theme
YOUR WORKSPACE AWAITS
Welcome back.
Sign in to your inventory workspace.
Username
Password
Show
Sign in to workspace →
Explore as guest (read-only, no password)
Demo accounts. Click to fill in:
Admin
Manager
Staff
← Back to home
StockFlow Inventory Suite · 2026


StockFlow
INVENTORY SUITE
Dashboard
INVENTORY
Products
Stock control
Movements
TRADE
Purchase orders
Sales orders
Suppliers
Customers
INSIGHTS
Reports
Activity log
API docs
ACCOUNT
My account
Users & roles
Aditi
Admin
Sign out →
StockFlow / Dashboard
Sat, 07 Nov 2026
18
Dark theme
YOUR DAILY PERSPECTIVE
Dashboard
Welcome back, Aditi. Here is how your inventory looks right now.
+ Sales order
+ Purchase order
Open stock control →
Stock value
₹128,198
At current catalog prices
Active products
30
32 total · 2 archived
Units on hand
284
Across active products
Low stock
13
At or below reorder level
Out of stock
5
Products with zero units
Open purchase orders
4
Awaiting delivery
Sales this month
₹5,582
1 order waiting to ship
LAST 30 DAYS
Stock in vs. out
Movement report →
2,000
1,800
1,600
1,400
1,200
1,000
800
600
400
200
0
09 Oct
13 Oct
17 Oct
21 Oct
25 Oct
29 Oct
02 Nov
06 Nov
Received
Issued
HEALTH
Stock levels
Healthy
Low stock
Out of stock
WHERE THE MONEY SITS
Stock value by category
Electronics
Home
Office
Hardware
Packaging
Cleaning
₹0
₹10,000
₹20,000
₹30,000
₹40,000
₹50,000
₹60,000
₹70,000
₹80,000
LAST 30 DAYS
Fastest-moving products
100
80
60
40
20
0
Demo Refuse Bag Pack
Demo Scrub Brush
Demo Hand Soap
Demo Microfiber Cloth
Demo Shipping Tape
NEEDS ATTENTION
Stock to review
View stock →


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Reports
Activity log
API docs
ACCOUNT
My account
StockFlow / Stock control
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Dark theme
EVERY UNIT ACCOUNTED FOR
Stock control
Receive stock, issue units, and monitor active products.
Manage products
Active products
30
Current catalog
Units on hand
284
Across active products
Low stock
13
Positive stock at or below threshold
Out of stock
5
Zero units available
Search
Name, SKU, or category
Stock level
Low stock
Apply
Reset
13 products
Active products only · Totals above cover all active products
Product
Available
Reorder level
Stock level
Actions
Demo Carton Box
DEMO-V1-023
5
5
Low stock
Receive Issue Count Reorder
Demo Desk Organizer
DEMO-V1-011
5
5
Low stock
Receive Issue Count Reorder
Demo File Folder
DEMO-V1-010
3
5
Low stock
Receive Issue Count Reorder


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ACCOUNT
My account
StockFlow / Purchase orders
Sat, 07 Nov 2026
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Dark theme
PURCHASE ORDER
PO-DEMO-003
Raised by demo_admin on 28 Oct 2026, 15:29
Back to purchase orders
STATUS
Partially received
40% received
Supplier
Demo Supply Partner 05
Product
Demo Power Bank
DEMO-V1-005
Ordered
30 units
Outstanding
18 units
Unit cost
₹929.38
Order value
₹27,881.40
Receive delivery
Download PDF
Cancel order
DELIVERIES
Receipts against this order
Time
Movement
Quantity
Document
By
31 Oct 2026, 15:29
Stock received
+12
DC-4102
demo_admin
Reverse

StockFlow Demo Traders

Bengaluru, Karnataka, India

TAX INVOICE

SO-DEMO-0001
Date: 24 Oct 2026
Status: Fulfilled

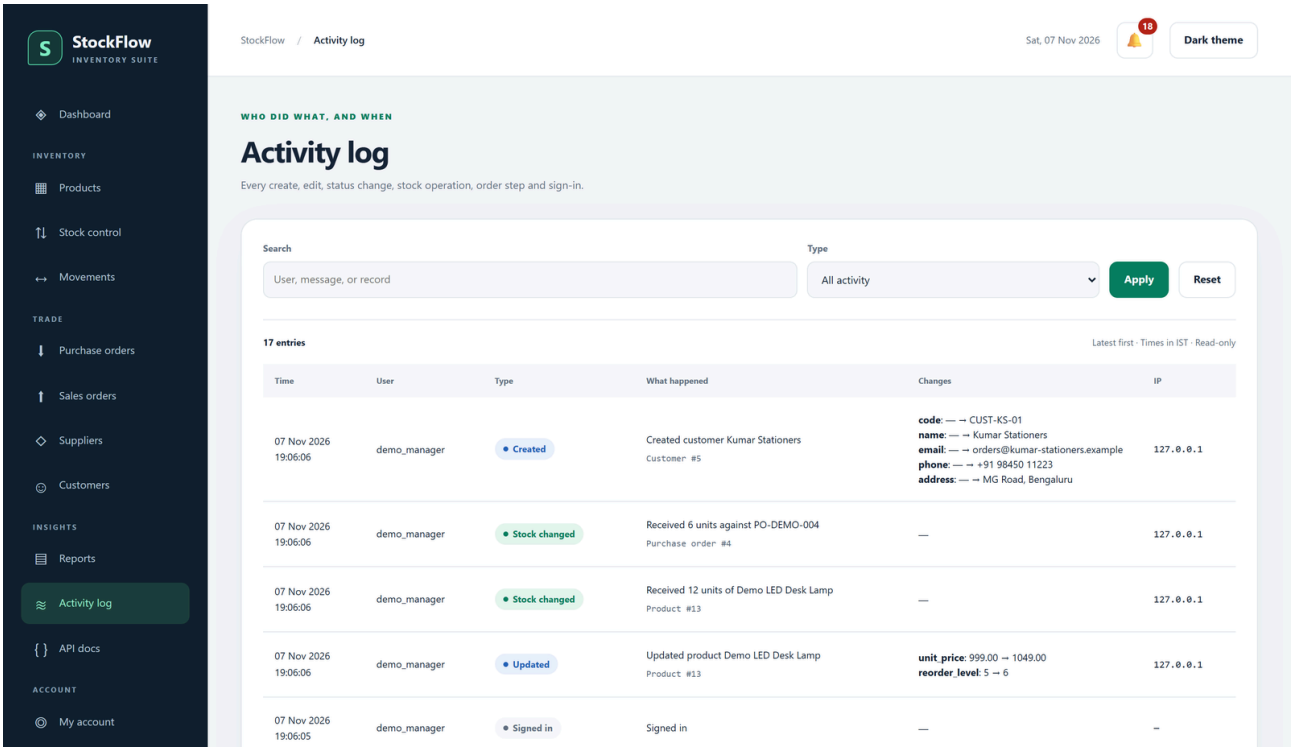
Bill to

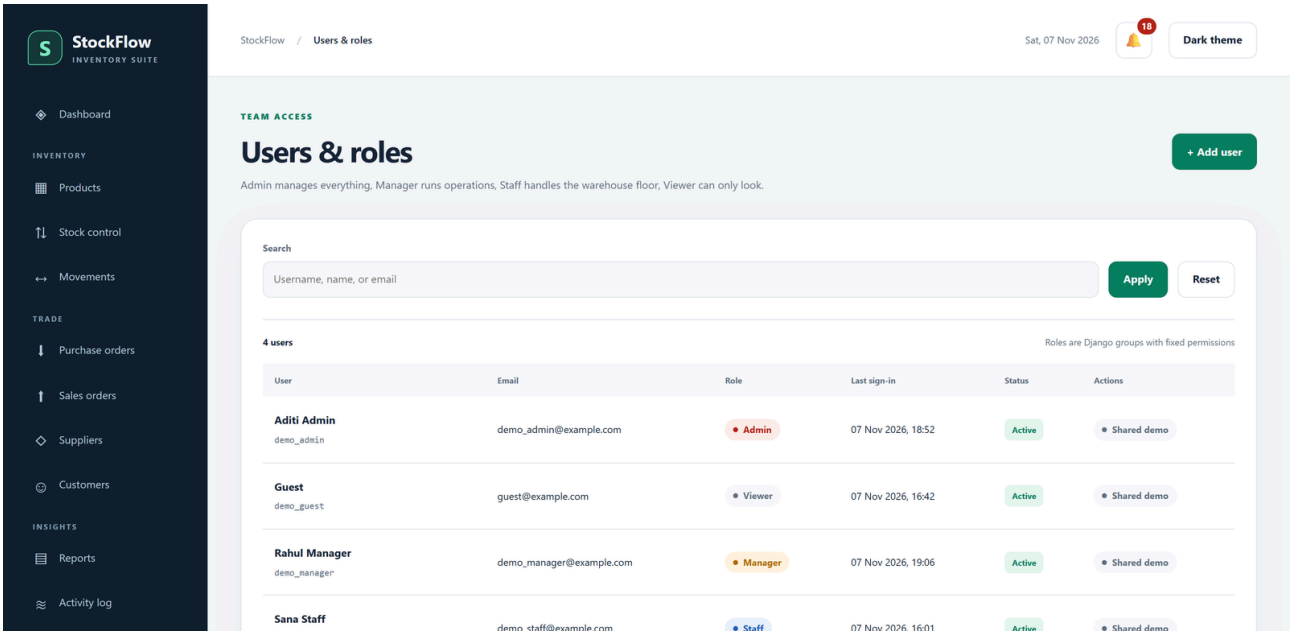
Sharma Electronics (DEMO-CUST-01)
Lajpat Nagar, New Delhi
orders@sharma-electronics.example +91 98100 00001

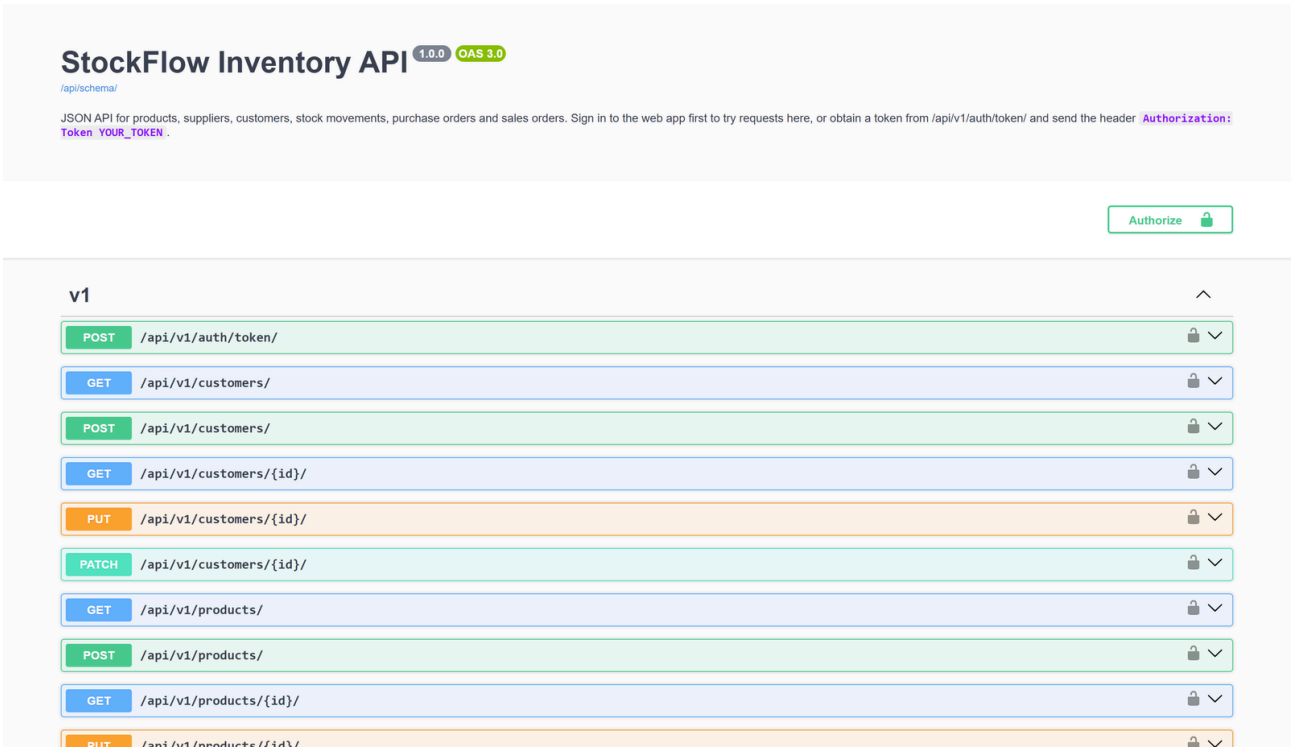
#	Item	Qty	Rate (INR)	Amount (INR)
1	Demo USB Hub DEMO-V1-004	4	899.00	3,596.00
2	Demo Ballpoint Pen Box DEMO-V1-008	3	120.00	360.00
			Total	**3,956.00**

Note: Demo order

Prices are as agreed on the order. This is a computer-generated document and needs no signature.




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Sat, 07 Nov 2026
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Dark theme
TEAM ACCESS
Users & roles
+ Add user
Admin manages everything, Manager runs operations, Staff handles the warehouse floor, Viewer can only look.
Search
Username, name, or email
Apply
Reset
4 users
Roles are Django groups with fixed permissions
User
Email
Role
Last sign-in
Status
Actions
Aditi Admin
demo_admin
demo_admin@example.com
Admin
07 Nov 2026, 18:52
Active
Shared demo
Guest
demo_guest
guest@example.com
Viewer
07 Nov 2026, 16:42
Active
Shared demo
Rahul Manager
demo_manager
demo_manager@example.com
Manager
07 Nov 2026, 19:06
Active
Shared demo
Sana Staff


StockFlow Inventory API
1.0.0
OAS 3.0
/api/schema/
JSON API for products, suppliers, customers, stock movements, purchase orders and sales orders. Sign in to the web app first to try requests here, or obtain a token from /api/v1/auth/token/ and send the header Authorization: Token YOUR_TOKEN .
Authorize
v1
POST /api/v1/auth/token/
GET /api/v1/customers/
POST /api/v1/customers/
GET /api/v1/customers/{id}/
PUT /api/v1/customers/{id}/
PATCH /api/v1/customers/{id}/
GET /api/v1/products/
POST /api/v1/products/
GET /api/v1/products/{id}/

6. Conclusion

StockFlow was successfully built in stages and deployed as a fully working web application. It meets its objective of giving a small business an accurate, real-time and accountable record of its stock, together with purchasing, sales, reporting and analytics, available from anywhere.

What was achieved

- A complete inventory, purchasing and sales system built with Django and PostgreSQL.
- Stock balances that stay correct under simultaneous use, with a full, permanent history.
- Role-based access, a read-only guest view and a complete activity log.
- A dynamic interface with live search, charts, PDF documents and a documented REST API.
- Automated testing on every change and a live deployment on free cloud hosting.

What I learned

- Designing a relational database and keeping data consistent with transactions and row locking.
- Building secure web applications: authentication, permissions, CSRF, HTTPS and protection against repeated sign-in attempts.
- Writing automated tests and running them in a continuous-integration pipeline.
- Building REST APIs and making pages dynamic without a separate frontend framework.
- Deploying with environment variables, managed databases and production security settings, and using Git for version control.

Future scope

- Barcode / QR-code scanning for receiving and issuing stock.
- Email or SMS alerts when products fall below their reorder level.
- Multiple warehouses or store locations with stock transfers.
- GST tax calculation on invoices and integration with accounting software.

7. Project Files Submission

The complete source code is available on GitHub (https://github.com/Md-Faiz-Alam/inventorymanagement-system) and the working system at https://stockflow-inventory-0qr7.onrender.com.

The submission file UGO24156735.zip contains:

Item	Contents
Source code	The full StockFlow project: Django apps (accounts, inventory, suppliers, sales, reports, audit, api), templates, static files, migrations, tests, requirements.txt, render.yaml and build.sh. Secret files (.env) are not included.
README.md	Features, local setup steps, testing commands and deployment guide.
Reports	Progress Report 1, Progress Report 2 and this Final Project Report (PDF).
Links	GitHub repository and live application URLs.